

Kundai Farai Sachikonye
Auf der Lichtung 19, 82178 Puchheim, Germany
github.com/fullscreen-triangle
kundai.sachikonye@bitspark.com
(+49) 1626386344

18 May 2026

Dr. Moritz Kleinert
Photonic Biosensors
Fraunhofer-Institut für Nachrichtentechnik — Heinrich-Hertz-Institut (HHI)
Berlin
Ref. 84447

Re: Wissenschaftliche*r Mitarbeiter*in für Biochemie und Oberflächenfunktionalisierung (Ref. 84447)

Dear Dr. Kleinert,

I am applying for the scientist position in biochemistry and surface functionalization at Fraunhofer HHI (Ref. 84447). My background combines formal biochemistry training and practical protein-handling experience with independent computational work on the optical physics and molecular recognition problems that underlie photonic biosensor performance. The surface chemistry specific to photonic integrated chips is the aspect I have not done on the bench; the biochemical foundation, the understanding of evanescent-field sensing mechanisms, and the analytical tools for biosensor signal processing are in place.

Biochemistry and assay development background. My BSc in Biochemistry and Cell Biology (Jacobs University Bremen) and MSc in Pharmaceutical Biotechnology (HAW Hamburg) together cover the full biochemical toolkit the position requires: protein purification and handling, antibody-based detection assays, bioconjugation strategies, ELISA development, and quality and reproducibility standards in bioanalytical method development. At the Universitätsklinikum Hamburg-Eppendorf (2017–2018) I developed automated LC-MS/MS and MALDI-TOF pipelines for proteomics and glycomics — work that required systematic protein immobilisation on MALDI targets, surface preparation, and strict protocol standardisation to achieve reproducible ion signals across sample series. At the University Medical Center Groningen (2013) I performed UPLC-MS/MS shotgun proteomics for cancer biomarker discovery, which required developing sample preparation and digestion protocols validated against a reference standard. These are the skills of assay development and experimental validation; the substrate here is PIC chip surfaces rather than MS plates, but the biochemical logic — controlled surface chemistry, selective capture, non-specific binding suppression, reproducibility at the device level — is the same.

Optical physics and photonic biosensing mechanisms. The detection mechanism in label-free photonic biosensing is a refractive index change in the evanescent field at the waveguide surface. Understanding where that signal comes from — and how large it should be for a given analyte-receptor interaction — requires first-principles optics. The Lagrangian platform I developed proves Beer-Lambert absorption, Kirchhoff circuit laws,

and thermal emission as formally identical operations via the Resonant Stack theorem: transmittance $T = 1/e = 0.3679$ is the universal fixed point in all three representations. This is exactly the physics of evanescent field coupling and absorption in a waveguide: the same partition-geometry argument that governs bulk absorption governs the sensing layer interaction. The Borgia framework derives electronic transitions, absorption, and emission spectra of molecules from bounded phase space geometry alone, with zero empirical parameters. Validated against NIST for nine elements and six molecules (self-consistency errors $<1.63\%$). For photonic biosensor design, this means I can predict the absorption cross-section and refractive index contribution of surface-bound molecules from molecular structure, before running the experiment — useful for pre-screening linker chemistries and bioreceptor candidates.

Molecular recognition and receptor–analyte binding. The quantitative output of a label-free biosensor is the binding kinetics and equilibrium K_d of the receptor–analyte interaction. The Crown Prince framework derives dissociation constants from bounded phase space geometry: K_d is determined by the partition structure of the receptor–ligand bounded oscillatory system, without empirical fitting. Validated against 39 NIST compounds. The Syndrome platform implements spectral DFT-based molecular recognition: three physicochemical channels (hydropathy, volume, charge) $\rightarrow$ Fourier transform $\rightarrow$ 36-dimensional cosine similarity ρ ; binding is predicted when $\rho \geq \rho^* = 0.72$ with $O(cL \log L)$ complexity. The mathematical structure is identical to what a photonic biosensor measures: spectral complementarity between receptor and analyte, with a threshold separating bound from unbound states. This gives me a computational layer for receptor design and binding affinity estimation that complements experimental validation on the chip.

Signal analysis and data evaluation. The Lavoisier framework processes spectral data at >100 GB scale: dual-pipeline numerical and CNN-based analysis (PyTorch), 98.9% NIST feature-extraction accuracy, Ray distributed computing. The signal analysis problem in photonic biosensing (extracting binding signal from interferometric or resonance-shift data against background noise and matrix interference) is structurally identical to the spectral peak extraction and ion annotation problem in mass spectrometry. I can therefore bring both analytical rigour and practical tooling to the data evaluation component of the role.

Background. I hold a BSc in Biochemistry and Cell Biology (Jacobs University Bremen), an MSc in Pharmaceutical Biotechnology (HAW Hamburg), and an MSc in Computational Biology (Universität Tübingen), and I conducted doctoral research in Computational Lipidomics at TUM. I am legally resident in Germany with full work authorisation and available immediately, including relocation to Berlin. English is native; German is conversational.

Yours sincerely,

Kundai Farai Sachikonye